LLM-AUGMENTED CONTRASTIVE LEARNING FOR MISINFORMATION DETECTION IN SOCIAL NETWORKS

Project report in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology

In Computer Science and Engineering (Internet of Things, cyber Security, BlockChain Technology)

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CERTIFICATE

This is to certify that the project titled **LLM-AUGMENTED CONTRASTIVE LEARNING FOR MISINFORMATION DETECTION IN SOCIAL NETWORKS** by the student of University of Engineering & Management, Kolkata who are mentioned above , in partial fulfillment of requirement for the degree of Bachelor of Computer Science and Engineering (Internet of Things, cyber Security, BlockChain Technology), is a bonafide work carried out by them under the supervision and guidance of **Prof. Sangita Dutta and Prof. Dr. Arijeet Ghosh** during **5th** Semester of academic session of 2025-2026. The content of this report has not been submitted to any other university or institute. I am glad to inform that the work is entirely original and its performance is found to be quite satisfactory.

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ABSTRACT

The rapid growth of digital platforms has led to an unprecedented spread of misinformation, posing serious challenges to societal stability, public trust, and informed decision-making. In recent years, Large Language Models (LLMs) and contrastive learning frameworks such as LLM-Augmented Contrastive Learning (LACL) have shown promising results in automated misinformation detection. However, existing approaches suffer from critical limitations, particularly in terms of emotional inconsistency in augmented data and the absence of reliable factual grounding. LLM-based augmentation techniques, while effective in preserving semantic meaning, often fail to retain the emotional tone of the original content, which is a key indicator in identifying misleading information. Additionally, reliance solely on textual patterns without external verification makes such systems vulnerable to hallucinations and factually incorrect predictions.

To address these challenges, this work proposes an enhanced LACL-based misinformation detection framework that integrates emotion-aware filtering and external fact verification within a unified pipeline. The proposed system incorporates an emotion-preserving filter that ensures augmented data maintains the original emotional characteristics by comparing emotion distributions and selectively retaining high-quality samples. Furthermore, a fact verification module is introduced to validate extracted claims against reliable external sources, thereby grounding the model's predictions in real-world knowledge. By combining semantic understanding, emotional consistency, and factual validation, the proposed approach significantly improves the robustness and reliability of misinformation detection. The overall system demonstrates improved representation learning and better generalization in identifying misleading content across diverse datasets. This integrated framework offers a scalable, interpretable, and practical solution for addressing the growing challenge of misinformation in modern digital ecosystems.

INTRODUCTION

The rapid growth of social media and digital communication platforms has significantly transformed the way information is generated and disseminated. While these platforms enable fast and widespread access to information, they have also contributed to the large-scale propagation of misinformation. False or misleading content, often driven by emotional appeal and rapid sharing, can influence public perception, create confusion, and lead to serious consequences in domains such as healthcare, politics, and public safety. As a result, the development of reliable automated systems for misinformation detection has become an essential area of research.

Recent advancements in artificial intelligence, particularly Large Language Models (LLMs), have enabled the creation of powerful text understanding systems capable of capturing semantic and contextual relationships within data. One such approach, LLM-Augmented Contrastive Learning (LACL), enhances model performance by generating augmented textual samples and learning meaningful representations through contrastive objectives. Although this approach improves data diversity and learning capability, it suffers from key limitations that affect its real-world effectiveness.

A major challenge in existing LACL-based systems is the loss of emotional consistency during data augmentation. Misinformation often relies on emotional expressions such as fear, urgency, or exaggeration to influence users. However, LLM-generated augmented data tends to preserve semantic meaning while altering or neutralizing emotional tone. This leads to inconsistencies between original and augmented samples, reducing the quality of contrastive learning and weakening the model's ability to detect emotionally driven misinformation. In addition, current systems lack mechanisms for verifying factual correctness and rely primarily on textual patterns, making them vulnerable to fluent but factually incorrect outputs generated by LLMs.

To address these issues, this work focuses on developing an enhanced misinformation detection framework that extends the LACL approach by incorporating emotion awareness and fact verification. The proposed system introduces an emotion-preserving filtering mechanism that evaluates the emotional similarity between original and augmented text and retains only those samples that maintain consistent emotional characteristics. This ensures higher-quality training data and improves the model's ability to capture both semantic and emotional features. Furthermore, a fact verification module is integrated into the framework to ground predictions in real-world knowledge by validating claims against reliable external sources.

The overall system follows a structured pipeline involving data preprocessing, LLM-based augmentation, emotion filtering, fact verification, and contrastive learning-based classification. By combining semantic understanding, emotional consistency, and factual validation, the proposed approach provides a more robust, interpretable, and scalable solution for misinformation detection in real-world scenarios.

Literature Survey

Hang Shen, Xiang Li, Xu Wang, Yuanfei Dai, Tianjing Wang, and Guangwei Bai, "LLM-Augmented Contrastive Learning for Misinformation Detection in Social Networks," IEEE Transactions on Computational Social Systems, vol. 13, no. 1, pp. 999–1014, 2026.

In "LLM-Augmented Contrastive Learning for Misinformation Detection in Social Networks" (IEEE TCSS, 2026), Shen et al. propose a novel framework that integrates Large Language Models (LLMs) with Contrastive Learning (CL) to address key challenges in misinformation detection, including semantic complexity, limited labeled data, and evolving rumor patterns. The framework introduces an LLM-driven data augmentation strategy that generates diverse and semantically consistent samples through prompt orchestration. These augmented samples are used to improve the discriminative power of the contrastive learning model. Additionally, the authors propose a contrastive loss-aware joint training mechanism, where the CL model guides the LLM to reduce hallucinations and improve data quality. Experimental results on multiple benchmark datasets demonstrate that the proposed method significantly outperforms traditional deep learning approaches and improves generalization across different social media platforms .

Y. Liu, Y. Wu, and J. Zhu, "Early Detection of Fake News on Social Media Through Propagation Path Classification," in Proc. AAAI Conference on Artificial Intelligence, 2018.

In "Early Detection of Fake News on Social Media Through Propagation Path Classification" (AAAI, 2018), Liu et al. focus on analyzing how misinformation spreads through social networks by modeling propagation paths. The proposed approach combines recurrent and convolutional neural networks to capture both temporal and structural patterns of information diffusion. By analyzing early-stage propagation behavior, the model aims to detect fake news before it spreads widely. While the method demonstrates improved early detection performance, it relies heavily on the availability of propagation data, which may not always be accessible in real-time scenarios, thereby limiting its applicability.

T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A Simple Framework for Contrastive Learning of Visual Representations," in Proc. International Conference on Machine Learning (ICML), 2020.

In "A Simple Framework for Contrastive Learning of Visual Representations" (ICML, 2020), Chen et al. introduce SimCLR, a foundational framework for contrastive learning that focuses on learning effective representations by maximizing similarity between augmented views of the same data sample. Although originally designed for visual data, the underlying principles of contrastive learning have been widely adopted in textual applications, including misinformation detection. The study highlights the importance of high-quality data augmentation for effective representation learning. However, the performance of the framework is highly dependent on the quality and consistency of augmented samples, which remains a challenge in text-based applications.

X. Feng, J. Qian, and Y. Li, "Bi-Level Graph Contrastive Learning for Rumor Detection," in Proc. IEEE International Conference on Data Mining (ICDM), 2022.

In "Bi-Level Graph Contrastive Learning for Rumor Detection" (ICDM, 2022), Feng et al. propose a graph-based contrastive learning approach that models both local and global structures of social networks. The framework leverages graph neural networks to capture propagation patterns and relationships between users and posts. By applying contrastive learning at multiple levels, the model improves feature representation and detection accuracy. However, the approach primarily focuses on structural information and often overlooks semantic and emotional aspects of textual content, which are crucial for identifying misinformation in real-world scenarios

X. Hu, Z. Zhang, and Y. Liu, "Leveraging Large Language Models for Fake News Detection via Multi-Perspective Reasoning," in Proc. ACM International Conference on Information and Knowledge Management (CIKM), 2023.

In "Leveraging Large Language Models for Fake News Detection via Multi-Perspective Reasoning" (CIKM, 2023), Hu et al. explore the use of Large Language Models as auxiliary reasoning agents rather than direct classifiers. The model uses LLMs to generate multiple perspectives and explanations for a given news item, which are then used to guide the detection process. This approach enhances interpretability and reasoning capability in misinformation detection. However, the lack of tight integration between the LLM and the detection model limits the effectiveness of knowledge transfer and Reduces overall performance consist

Research Gap

Despite the effectiveness of the LLM-Augmented Contrastive Learning (LACL) framework in misinformation detection, several critical limitations remain that impact its performance and reliability in real-world applications. One of the primary challenges is the loss of emotional consistency during LLM-based data augmentation. In the original LACL framework, Large Language Models are used to generate augmented textual samples that are semantically similar to the original input. However, while these models preserve semantic meaning, they often fail to retain the emotional tone and intensity of the original text. This becomes a significant issue in misinformation detection, as emotional signals such as fear, anger, urgency, and exaggeration are strong indicators of misleading or false information. For instance, an emotionally intense statement may be transformed into a neutral sentence during augmentation. Although both versions may convey similar meanings, the loss of emotional cues creates inconsistencies between original and augmented samples. This leads to the formation of poor-quality positive pairs in contrastive learning, where emotionally dissimilar samples are incorrectly treated as similar, ultimately weakening the model's representation learning capability.

To address this limitation, the proposed work introduces an emotion-preserving filtering mechanism. This module extracts emotion distributions from both original and augmented texts and evaluates their similarity using cosine similarity. Only those augmented samples that maintain a sufficient level of emotional consistency are retained for training, while others are discarded. In addition, a dynamic thresholding strategy based on emotional intensity is applied to ensure context-aware filtering. This approach improves the quality of the augmented dataset and ensures that both semantic and emotional features are preserved, thereby enhancing the effectiveness of contrastive learning.

Another significant gap in the existing LACL framework is the lack of external fact verification. Current approaches rely heavily on LLM-generated content and textual patterns, which are often prone to hallucinations, where the model produces fluent but factually incorrect information. As a result, the system may incorrectly associate strong emotional expressions with misinformation without validating factual correctness. For example, a statement may appear highly emotional and persuasive while being factually false, or vice versa. The absence of a mechanism to verify claims against reliable external sources reduces the model's ability to generalize and undermines its trustworthiness.

To overcome this issue, the proposed framework incorporates a fact verification module that introduces factual grounding into the detection process. This module extracts key claims from the input text and retrieves relevant information from trusted external sources such as Wikipedia. A similarity score is then computed between the extracted claim and the verified information to determine factual consistency, which is integrated with the model's textual representations. This enables the system to consider both linguistic features and factual accuracy during prediction, making it more robust and reliable.

By addressing the issues of emotional inconsistency and lack of factual grounding, the proposed system enhances the traditional LACL framework and transforms it into a more comprehensive and trustworthy solution for real-world misinformation detection.

PROPOSED METHODOLOGY

RESULT & ANALYSIS

Conclusion and Future Scope

This work presents an enhanced framework for misinformation detection by extending the LLM-Augmented Contrastive Learning (LACL) approach with emotion-aware filtering and external fact verification. While the original LACL framework effectively leverages Large Language Models for data augmentation and contrastive learning for representation learning, it suffers from important limitations such as the loss of emotional consistency and the absence of factual grounding. These challenges significantly affect the quality of augmented data and reduce the reliability of predictions, particularly in real-world scenarios where misinformation often relies on emotional manipulation and misleading claims. To overcome these issues, the proposed system introduces an emotion-preserving filtering mechanism that ensures augmented samples retain the emotional tone and intensity of the original text. By extracting emotion distributions and applying similarity-based filtering along with dynamic thresholding, the system improves the quality of contrastive learning pairs and enables the model to capture both semantic and emotional features more effectively.

In addition to addressing emotional inconsistency, the proposed framework incorporates a fact verification module that grounds model predictions in real-world knowledge. This module extracts claims from input text and retrieves relevant information from reliable external sources, such as Wikipedia, to evaluate factual consistency. By integrating this fact consistency score with textual representations, the system ensures that predictions are not solely based on linguistic patterns or emotional cues but are also supported by verified evidence. The combination of semantic understanding, emotional awareness, and factual validation results in a more robust, accurate, and trustworthy misinformation detection system. Overall, the proposed approach significantly enhances data quality, improves representation learning, and increases the model's ability to generalize across different datasets and real-world contexts.

Despite these improvements, there remains considerable scope for future enhancement of the proposed framework. One important direction is the extension of the system to handle multimodal data, including images, videos, and audio, as misinformation on modern platforms is often presented in multiple formats. Incorporating multimodal analysis can provide a more comprehensive understanding of misleading content and further improve detection performance. Additionally, future work can explore the use of more advanced and domain-specific Large Language Models that are fine-tuned for misinformation detection tasks, enabling better context understanding and more reliable data augmentation. Expanding the fact verification component to include multiple knowledge bases and real-time fact-checking mechanisms can further strengthen the system's ability to handle dynamic and evolving information.

Furthermore, improving computational efficiency and optimizing the model for large-scale deployment will be crucial for real-time applications in social media monitoring systems. Enhancing interpretability by providing explanations for model predictions can also increase transparency and user trust, which is essential in sensitive domains such as misinformation detection. Overall, the proposed framework lays a strong foundation for future research by addressing key limitations of existing methods and offering a scalable and adaptable solution. With further advancements, the system has the potential to evolve into a comprehensive and intelligent platform capable of effectively combating misinformation in increasingly complex digital environments.

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